28 October 2026

Dear Editors,

Article type. We submit our manuscript, "Analysis-unit sensitivity distinguishes donor-level expression enrichment from within-donor coupling in human kidney thick ascending limb states", for consideration as a Research Article.

The question. Single-cell kidney studies commonly describe a disease-associated state as a property of a cell type, although donor composition, local neighborhood structure, and within-donor coupling answer different questions. We asked whether these analysis units lead to the same subtype-level conclusion in the human thick ascending limb (TAL).

What we found. Using 106,851 KPMP single-nucleus TAL cells, we compared donor pseudobulk, Milo neighborhood abundance, neighborhood-expression association, and within-donor cell-rank association across four TAL subtypes. The units were not interchangeable. In aTAL2, donor-level and neighborhood-expression directions were positive, whereas the within-donor association was negative in combined disease and AKI (FDR = 0.0086 for each). Complete and subset pseudobulk directions agreed in 48 of 60 comparisons, but 12 changed direction. The analysis unit is therefore not a neutral technical choice; it changes which biological quantity is being estimated. A donor-level decomposition showed that the overall PAPPA2 difference was mainly a subtype-expression effect (+0.641) rather than a subtype-composition effect (+0.010). In a separate prespecified 400-gene panel, donor-versus-within reversal occurred in 18.6% of target genes versus 12.7% of expression-matched random sets (empirical P = 0.000999).

Independent support. We tested the key directions in two non-KPMP kidney resources. GSE211785 supported donor-level PAPPA2 disease elevation, and GSE254185 showed positive donor-level and neighborhood-abundance directions alongside negative neighborhood-expression and within-donor directions in injured TAL. A donor-stratified signature transfer achieved 93.9% accuracy among assigned held-out cells and 93.9% aTAL2 recall. The external data support a donor-level direction and the general analysis-unit problem, but do not provide exact aTAL2 replication.

Kidney regulatory context. In NephQTL2, PAPPA2 was a strong tubulointerstitial eGene (FDR = 1.64e-14) without glomerular eQTL support. We also provide a prespecified boundary analysis: PAPPA2 was not among FDR-significant PTWAS genes, and same-cohort snATAC gene activity did not show higher disease accessibility. We treat this as regulatory context rather than genetic validation and state that boundary explicitly.

Why Human Genetics. The manuscript connects human kidney single-cell genomics, donor-aware statistical inference, external cohort transfer, and kidney eQTL context. Its contribution is not a new algorithm or a claim that PAPPA2 causes disease. It is a rigorous demonstration that the unit of inference changes subtype-level conclusions and should be reported explicitly. We believe this fits the journal's readership in human genetics, functional genomics, and disease mechanism.

Declarations. The manuscript is original, has not been published, and is not under consideration elsewhere. All authors have read and approved the submission. The authors declare no competing interests and no specific funding. The study used de-identified public or controlled-access data; no new human samples were collected. No clinical trial was performed. The manuscript is not intended for a special issue. Because we have no publication fund, we request consideration under the traditional subscription route and do not require Open Choice.

We have not requested the exclusion of any reviewer.

Thank you for considering our manuscript.

Yours sincerely,

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On behalf of all authors